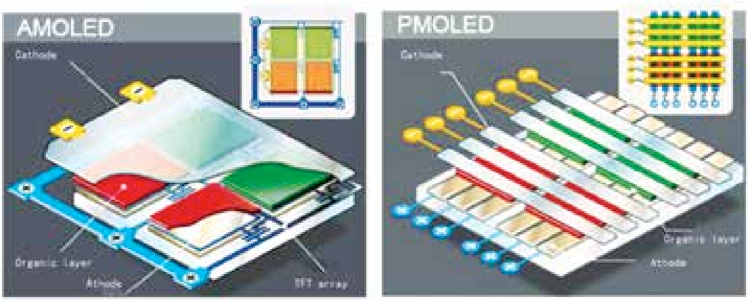
1. Passive Matrix OLEDs (PMOLEDs): In a typical PMOLED, you would find long strips of cathode, organic layers and strips of anode with the anode strips being perpendicular to the cathode strips where at the intersection make up the pixels where holes are filled i.e. the light is emitted. The entire external circuitry applies current to selected strips of anode and cathode to control which pixels glow and which pixels remain off. The amount of current applied is directly proportional to the brightness the pixels in an OLED panel will attain and thus, PMOLEDs generally consume more power than other types of OLED, mainly due to the power needed for the external circuitry. The requirement of high voltage in the circuit also affects the lifespan of the PMOLED panel.

PMOLEDs are preferred for display of text and small icons and are best suited for smaller screen sizes such as small cell phones, MP3 players, Smart watches, Flexible wearable bands etc.

2. Active-matrix OLEDs (AMOLEDs): The word AMOLED reminds of Samsung and its gorgeous displays with great contrast levels, often seen on its flagship phones. Full layers of cathode, organic molecules and anode with an overlay of a thin film transistor (TFT) array forming a matrix is the assembly, typically seen in an AMOLED. The TFT array controls which pixels get turned on to form the image.

As the TFT array requires less power than external circuitry, AMOLEDs consume less power than PMOLEDs and thus are preferred for large display panels like that in a laptop or a television. AMOLEDs, also have faster refresh rates, suitable for video playback. The best uses for AMOLEDs are high-end computer monitors and laptop display panels, large-screen



AMOLED and PMOLED - Most commonly used OLED types.